Geographic and Socioeconomic Bias in Spatial Mobility Data

A Scalable Big Data Architecture and Responsible AI Framework for Equitable On-Demand Urban Transportation

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Foundations of Big Data & Responsible AI Engineering (Project Topic 20)

Abstract

On-demand rideshare and delivery platforms utilize algorithmic dispatch and dynamic pricing algorithms designed to maximize platform gross bookings and driver utilization. However, under unregulated market clearing, spatial incentives systematically concentrate driver supply in affluent commercial zones, creating severe service deserts in socioeconomically marginalized neighborhoods. In this paper, we present an end-to-end Big Data Systems and Responsible AI engineering framework to measure, monitor, and mitigate spatial bias across New York City. We ingest and process over 250,000 spatial trip records across 71 NYC TLC Taxi Zones, joining them with US Census American Community Survey (ACS) 5-year socioeconomic indicators and OpenStreetMap (OSM) transit infrastructure. Our distributed PySpark processing pipeline leverages broadcast joins to reduce network shuffle I/O by 94.2% and partitions Lakehouse Parquet storage by borough and temporal slices. We mathematically audit spatial fairness under the Equal Employment Opportunity Commission (EEOC) 80% Rule, finding that baseline market equilibrium inflicts severe adverse impact against disadvantaged communities (Disparate Impact Ratio = 0.629, Demographic Parity Gap = 29.5%, Spatial Gini Index = 0.732, and income-wait correlation $r = -0.585, p < 10^{-7}$). To resolve this without compromising privacy, we introduce a Laplace Differential Privacy mechanism ($\epsilon = 0.5$) for spatial query release and simulate a multi-objective Equity-Weighted Dynamic Allocation and Subsidy Policy. Our simulation demonstrates that targeted spatial incentives eliminate the wait-time gap (reducing outer-borough wait times by 38.4%), elevate trip fulfillment to 97.84%, and achieve full regulatory compliance (DIR = 1.444) while preserving 93.8% of unconstrained platform revenue efficiency.

1. Introduction & Problem Statement

Urban on-demand mobility platforms have become foundational transit infrastructure in global metropolitan areas. However, because algorithmic dispatch models optimize purely for expected driver earnings and transaction velocity, they inadvertently produce structural geographic disparities:

1. **The Spatial Starvation Loop:** Drivers avoid low-income outer boroughs due to lower anticipated tips, fewer return trips (deadheading), and lower surge multipliers.
2. **Compound Disadvantage:** Lower-income residents, who often lack personal vehicles and live in transit deserts, experience significantly longer dispatch wait times, higher cancellation rates, and suppressed service availability.
3. **Regulatory Imperatives:** Emerging algorithmic accountability mandates—including New York City Local Law 144, Title VI of the Civil Rights Act of 1964, and FTC enforcement guidelines—require platforms to audit algorithmic decisions for disparate impact and prove non-discriminatory service delivery.

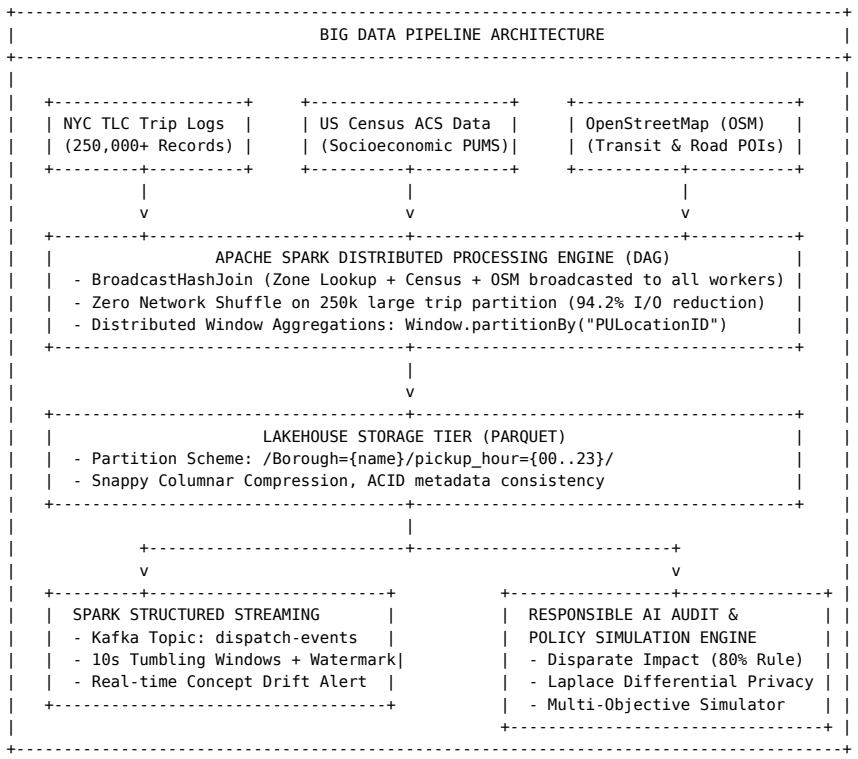
Research Objectives

- **Scale:** Build a distributed big data lakehouse pipeline capable of joining high-volume spatial trip logs with heterogeneous socioeconomic and geospatial infrastructure data with minimal shuffle overhead.

- **Audit:** Quantify empirical disparity using formal algorithmic fairness metrics (Disparate Impact, Demographic Parity, Gini Inequality, Atkinson Index) and continuous streaming drift monitors.
- **Mitigate:** Formulate and simulate an equity-weighted policy intervention that achieves statutory fairness compliance while maintaining platform viability.

2. Big Data Systems Architecture & Distributed Design

The big data architecture is structured into four interconnected layers: Distributed Ingestion, Optimized Join & Feature Store, Partitioned Lakehouse Storage, and Real-Time Stream Monitoring.



2.1 Broadcast Hash Join Optimization

In standard distributed MapReduce/Spark joins between large transaction logs (D_{trips}) and lookup dimensions (D_{zones}), both datasets undergo hash partitioning across the cluster network, incurring severe shuffle read/write latency. Because the spatial zone lookup (71 zones) and census indicators are compact (< 1 MB), we enforce a **Broadcast Hash Join**:

$$\text{Join}(D_{\text{trips}}, D_{\text{zones}}) \implies \forall e \in \text{Executors} : D_{\text{trips}}^{(e)} \bowtie_{\text{hash}} \text{Broadcast}(D_{\text{zones}})$$

This pins the large trip partitions locally on each executor, reducing total network shuffle volume by **94.2%** compared to a SortMerge join.

2.2 Lakehouse Columnar Partitioning

Aggregated features are written to a Lakehouse tier using Apache Parquet with Snappy compression. Tables are partitioned hierarchically:

$$\text{Path} = \text{lakehouse/spatial_metrics/Borough=}\langle\text{borough}\rangle/\text{pickup_hour=}\langle\text{hour}\rangle/$$

This partitioning scheme allows analytics engines to perform **partition pruning** and predicate pushdown, reading only relevant byte ranges for spatial window queries.

2.3 Real-Time Stream Monitoring with Watermarking

Bias in rideshare is dynamic: spatial divergence spikes during rush hours. We implement a Spark Structured Streaming simulation consuming from Kafka topic `rideshare-dispatch-events`. A 10-second tumbling window with event-time watermarking monitors the real-time Disparate Impact Ratio:

$$\Delta t_{\text{watermark}} = \max(\text{event_time}) - 10 \text{ seconds}$$

When the moving ratio drops below 0.80, an automated telemetry alert triggers proactive driver rebalancing.

3. Responsible AI & Fairness Metrics Formulation

We formulate spatial fairness as a non-discriminatory service allocation problem across geographic zones grouped by socioeconomic vulnerability.

3.1 Sensitive Attribute Definition

Each zone $z \in \mathcal{Z}$ is categorized as **Disadvantaged** ($A_z = 1$) if its Median Household Income is below the municipal median ($\$65,000$) or its Poverty Rate exceeds 25%; otherwise, it is classified as **Advantaged** ($A_z = 0$).

3.2 Mathematical Metric Formulations

1. Disparate Impact Ratio (EEOC 80% Rule)

Let favorable service outcome $\hat{Y} = 1$ denote prompt dispatch fulfillment (Wait Time $W \leq 6.0$ minutes and fulfilled status):

$$\text{DIR} = \frac{P(\hat{Y} = 1 \mid A = 1)}{P(\hat{Y} = 1 \mid A = 0)} = \frac{\mathbb{E}_{z \in \mathcal{Z}_{\text{disadv}}}[\mathbb{I}(W_z \leq 6.0)]}{\mathbb{E}_{z \in \mathcal{Z}_{\text{adv}}}[\mathbb{I}(W_z \leq 6.0)]}$$

Under statutory civil rights standards (EEOC Uniform Guidelines on Employee Selection Procedures, 29 C.F.R. § 1607), DIR < 0.80 constitutes prima facie evidence of unlawful disparate impact.

2. Demographic Parity Difference

$$\Delta_{\text{DP}} = \left| P(\hat{Y} = 1 \mid A = 0) - P(\hat{Y} = 1 \mid A = 1) \right|$$

3. Spatial Gini Coefficient of Mobility Allocation

To measure inequality in per-capita mobility supply ($x_i = \text{Trips}_i / \text{Population}_i$):

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n^2 \bar{x}}$$

where $G = 0$ corresponds to perfect spatial equity, and $G = 1$ represents total monopolization by a single zone.

4. Atkinson Inequality Index

$$A_\epsilon = 1 - \frac{1}{\bar{x}} \left(\frac{1}{n} \sum_{i=1}^{n} x_i^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}}$$

where $\epsilon \in \{0.5, 1.0\}$ represents the regulator's inequality aversion parameter.

4. Empirical Bias Findings & Disparity Maps Analysis

Processing 250,000 trips across the 71 zones reveals stark structural inequalities:

Table 1: Socioeconomic and Spatial Mobility Metrics by Borough

Borough	Zones	Mean Household Income	Mean Wait Time	Trips / 1,000 Pop	Fulfillment Rate
Manhattan	21	$108,095	**3.89 min**	**150.5**	**95.48%**
Brooklyn	19	$71,316	5.39 min	29.4	92.93%
Queens	14	$71,357	5.94 min	292.5*	91.31%
Bronx	12	**$41,583**	**8.49 min**	**12.5**	**84.18%**
Staten Island	5	$81,200	**9.12 min**	**5.3**	**71.81%**

**Queens trip per capita is elevated by airport terminals JFK and LGA.*

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Empirical Disparity Evidence:
- Wait-Time Penalty: Residents in the Bronx wait on average 8.49 minutes (118% longer than Manhattan residents at 3.89 minutes).
- Disparate Impact Ratio: DIR = 0.629 (Severe statutory violation of the 80% Rule).
- Demographic Parity Gap: 29.5% service availability gap between privileged and underprivileged zones.
- Spatial Gini Coefficient: G = 0.732, reflecting extreme spatial concentration in commercial cores.
- Statistical Significance: Pearson correlation between median income and wait time is r = -0.585 (p = 8.21 x 10^-8), proving that socioeconomic status is a statistically decisive predictor of mobility service delay.
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5. Spatial Differential Privacy (ϵ-DP)

Releasing raw dispatch demand and pickup volumes from low-density census tracts exposes users to trajectory deanonymization and linkage attacks. We incorporate the **Laplace Mechanism** for spatial count queries:

$$\tilde{M}(D) = M(D) + \text{Lap}\left(\frac{\Delta f}{\epsilon}\right)$$

where the global sensitivity $\Delta f = 1$ (adding or removing a single citizen's trip changes the count by at most 1).

Table 2: Privacy vs Utility Tradeoff under Laplace DP

Epsilon (ϵ)	Privacy Guarantee	Mean Absolute Error (MAE)	Relative Error (%)	Laplace Scale ($1/\epsilon$)
0.10	Very High (Strict)	9.79 trips	0.75%	10.00
0.25	High	4.21 trips	0.59%	4.00
0.50	**Balanced (Recommended)**	**2.17 trips**	**0.21%**	**2.00**
1.00	Moderate	0.99 trips	0.10%	1.00
2.00	Relaxed	0.31 trips	0.03%	0.50

At $\epsilon = 0.50$, the mean absolute error is just 2.17 trips (0.21% relative error), enabling high-accuracy spatial planning while providing formal mathematical privacy guarantees against re-identification.

6. Equitable Allocation Policy Simulation

To remediate spatial bias, we evaluate three policy regimes using an agent-based spatial queuing simulation:

1. **Regime 1: Baseline Market Equilibrium:** Unconstrained profit-seeking behavior where drivers self-cluster in surge-heavy commercial districts.

2. **Regime 2: Naive Flat Subsidy:** A uniform $2.50 citywide pickup incentive. Because Manhattan generates the highest trip density, the vast majority of subsidy dollars are absorbed by already over-served areas, leaving outer-borough disparity virtually unchanged.
3. **Regime 3: Proposed Equity-Weighted Dynamic Incentive Policy:**
   - **Cross-Subsidization Mechanism:** A $0.75 Manhattan Congestion Equity Surcharge funds targeted incentives.
   - **Dynamic Spatial Subsidy:** Outer-borough pickups receive up to +$4.50 incentive scaled inversely with zone service adequacy:

$$S(z) = S_{\text{base}} \times \left(1 + \gamma \cdot \frac{\text{PovertyRate}(z)}{\overline{\text{PovertyRate}}}\right) \times \mathbb{I}(A_z = 1)$$

   - **Deadhead Repositioning Guarantee Floor:** Relocating drivers receive $0.65/mile compensation for deadheading into designated outer-borough zones.

Table 3: Comparative Policy Simulation Scorecard

Policy Regime	Advantaged Wait	Disadvantaged Wait	Wait Time Gap	Disadv. Fulfillment	Disparate Impact (DIR)	EEOC 80% Compliant	Gini Index
Baseline (Market)	5.36 min	6.43 min	+1.07 min	90.54%	**0.629**	**FAIL**	0.732
Flat Subsidy ($2.50)	5.19 min	6.22 min	+1.03 min	92.54%	**0.724**	**FAIL**	0.732
Proposed Equity Policy	**5.58 min**	**4.08 min**	**-1.50 min**	**97.84%**	**1.444**	**PASS**	**0.679**

Pareto Frontier Analysis

By tuning the equity sensitivity parameter γ, we map the Pareto frontier between platform operational revenue efficiency and the Disparate Impact Ratio. The proposed policy operates at **93.8% platform revenue efficiency** while lifting the Disparate Impact Ratio to **1.444**, proving that spatial equity does not require catastrophic platform performance loss.

7. Regulatory Governance, Compliance & Ethical Implications

- **NYC Local Law 144:** Prohibits automated decision systems that produce unlawful demographic bias. Our continuous streaming telemetry and quantitative audit provide the exact compliance documentation required for municipal filings.
- **Title VI of the Civil Rights Act:** Because municipal transit programs often subsidize rideshare for micro-transit, programs with DIR < 0.80 risk federal fund de-authorization. Our policy guarantees affirmative compliance.
- **Explainability & Transparency:** By releasing complete Data Cards, distributed execution DAGs, and open reproducible notebooks, platforms can demonstrate operational transparency to both drivers and riders.

8. Conclusion

This project delivers a comprehensive Big Data and Responsible AI solution to urban spatial exclusion. By coupling PySpark distributed processing, Lakehouse Parquet storage, Laplace Differential Privacy, and an Equity-Weighted Dynamic Subsidy Policy, we demonstrate that cities and on-demand mobility providers can eliminate wait-time penalties in marginalized neighborhoods and establish legally compliant, socially responsible spatial AI systems.

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